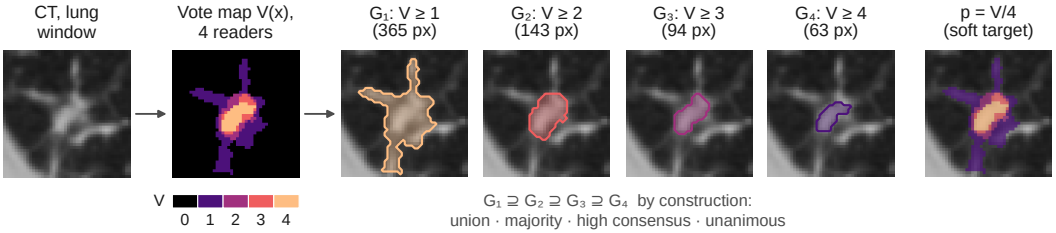
a One vote map, four nested references



b Thirteen trained configurations; each differs from base arm B only in the shaded cells

	Dice term	Pixel term	Encoder	Transl.	Backgr.	Runs	CV	Test	Offset	FOV
Supervision target										
A union	**G_1**	**G_1**	R34			5	●	●	●	○
B majority (base)	G_2	G_2	R34			5	●	●	●	●
C majority + soft	G_2	**p**	R34			5	●	●	●	○
D soft only	**p**	**p**	R34			5	●	●	●	○
E SVLS	G_2	**SVLS(G_2)**	R34			5	●	●	●	○
F SoftSeg*	**—**	**p (AW)**	R34			5	●	●	●	○
Architecture (target B)										
ResNet-18	G_2	G_2	**R18**			5	●	●	●	○
ConvNeXt-Tiny	G_2	G_2	**CNX-T**			5	●	●	●	○
PVT v2-b1	G_2	G_2	**PVT**			5	●	●	●	○
Plain U-Net	G_2	G_2	**none**			5	●	●	●	○
Data protocol (target B)										
+ translation ±32 px	G_2	G_2	R34	●		5	●	●	●	●
+ background 50%	G_2	G_2	R34		●	2	●	●	○	●
+ both	G_2	G_2	R34	●	●	2	●	●	○	●
Noise floor										
A, B × 3 seeds			R34			12	●	○	○	○

Shaded: where a configuration differs from base arm B. R34, ImageNet ResNet-34 encoder; p = V/4; AW, Adaptive Wing loss. *F changes target, activation and loss together and is reported outside the factorial design. Dots at right: where each configuration was evaluated. 71 LIDC training runs; all share optimiser, schedule, splits and the Dice + BCE objective.

c One forward pass, every protocol

References	× 4 (V ≥ 1 … V ≥ 4)
Thresholds	× 33, 0.001–0.999
Operating points	fixed 0.5 · calibrated on validation · area-matched to G_2
Extent	patch · windows 128–512 · slice · volume (× 8)
Offset	d = 0–56 px × 6 directions

d Data, splits and replication

LIDC-IDRI: 1018 series parsed, 1010 enter the study; case-level split; lesion = V ≥ 2 component ≥ 3 mm

- Cross-validation: 808 cases, 5 folds, 634 contributing, 8471 instances
- Held-out test: 202 cases, 158 contributing, 2062 instances

thresholds selected on validation folds, imported unchanged

External: QUBIQ 2021, 7 tasks from 4 collections (brain growth, brain tumour × 3, kidney, prostate × 2; 3–7 raters; CT and MRI) × 3 targets × 5 folds = 105 runs